MATH 152
Mrs. Bonny Tighe

QUIZ 5

25 points
7.7

NAME Solution

TA_____ Wed 10/9/2019

1. Find the error bounds for all three methods. $\int_1^2 \sqrt{x}\,dx, \quad n = 4$

Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then $|E_T| \le \dfrac{K(b-a)^3}{12n^2}$ $\quad |E_M| \le \dfrac{K(b-a)^3}{24n^2}$

Suppose $|f''''(x)| \le K$ *for* $a \le x \le b$ then $|E_S| \le \dfrac{K(b-a)^5}{180n^4}$

$f(x) = x^{1/2}$

$f'(x) = \frac{1}{2}x^{-1/2}$

$f''(x) = -\frac{1}{4}x^{-3/2} = -\frac{1}{4x^{3/2}}$ largest when $x=1$

$K = \left|\frac{1}{4}\right|$

$f'''(x) = \frac{3}{8}x^{-5/2}$

$f''''(x) = -\frac{15}{16}x^{-7/2}$

$\frac{-15}{16x^{7/2}}$

largest when $x = 1$

$K = \left|\frac{15}{16}\right|$

$|E_T| \le \dfrac{\frac{1}{4}(2-1)^3}{12(4)^2} = \boxed{\dfrac{1}{4(12)(16)}}$

$|E_M| \le \dfrac{\frac{1}{4}(2-1)^3}{24(4^2)} = \boxed{\dfrac{1}{4(24)(16)}}$

$E_S \le \dfrac{\frac{15}{16}(2-1)^5}{180(4)^4} = \boxed{\dfrac{15}{16(180)(4^4)}}$

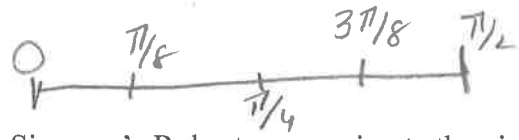
2. A radar gun was used to record the speed of a runner during the first 5 seconds of a race (see the table). Use Simpson's Rule to estimate the distance the runner covered during those 5 seconds.

t (s)	v (m/s)	t (s)	v (m/s)
0	0	3.0	12
0.5	4	3.5	10
1.0	7	4.0	11
1.5	8	4.5	10
2.0	9	5.0	9
2.5	11		

$\Delta x = 1/2$

$S_{10} = \frac{1/2}{3}\left[0 + 4(4) + 2(7) + 4(8) + 2(9) + 4(11) + 2(12) + 4(10) + 2(11) + 4(10) + 9\right]$

$\Delta x = \pi/8$



3. Use (a) the Trapezoidal Rule and (b) Simpson's Rules to approximate the given integral with the specified value of n for $\int_0^{\pi/2} \sin 2\theta \, d\theta, \quad n = 4$

$$T_4 = \frac{\pi/8}{2}\left[\sin 0 + 2\sin \pi/4 + 2\sin \pi/2 + 2\sin 3\pi/8 + \sin \pi/2\right]$$

$$= \frac{\pi}{16}\left[0 + 2(\sqrt{2}/2) + 2(1) + 2(\sqrt{2}/2) + 0\right]$$

$$= \frac{\pi}{16}\left[2 + 2\sqrt{2}\right]$$

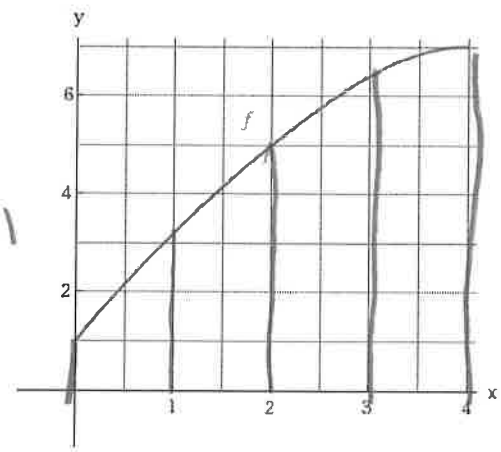
$$S_4 = \frac{\pi/8}{3}\left[\sin 0 + 4\sin \pi/4 + 2\sin \pi/2 + 4\sin 3\pi/8 + \sin \pi\right]$$

$$= \frac{\pi}{24}\left[0 + 4(\sqrt{2}/2) + 2(1) + 4(\sqrt{2}/2) + 0\right]$$

$$= \pi/24\left[2 + 4\sqrt{2}\right]$$

4. $I = \int_0^4 f(x)dx$ where f is the function whose graph is shown.

$\Delta x = 1$



$M_4 = 1[2 + 4 + 6 + 7]$

$L_4 = 1[1 + 3 + 5 + 6.5]$

$R_4 = 1[3 + 5 + 6.5 + 7]$

(a) Use the graph to approximate M_4, L_4, R_4

5. How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^2 \frac{1}{x^2}dx$ ~~$\int_1^2 \frac{1}{\sqrt{x}}dx$~~ is accurate to within 0.001?

$f(x) = x^{-2}$

$f'(x) = -2x^{-3}$

$f''(x) = 6x^{-4}$

$K = 6$

$\frac{6}{x^4}$ largest when $x = 1$

$$\frac{6(2-1)^3}{24n^2} \le \frac{1}{10^3}$$

$$\frac{10^3(6)}{24} \le n^2$$

$$n \ge \sqrt{\frac{10^3}{4}}$$

the next largest integer

MATH 152 **QUIZ 5A** NAME Solution

Mrs. Bonny Tighe 25 points

7.7 TA_____ Wed 10/9/2010

1. Find the error bounds for all three methods. $\int_2^3 \frac{1}{x}dx, \quad n=4$

Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then $|E_T| \le \frac{K(b-a)^3}{12n^2}$ $|E_M| \le \frac{K(b-a)^3}{24n^2}$

Suppose $|f''''(x)| \le K$ *for* $a \le x \le b$ then $|E_S| \le \frac{K(b-a)^5}{180n^4}$

$f(x) = x^{-1}$

$f'(x) = -x^{-2}$

$f''(x) = 2x^{-3} = \frac{2}{x^3}$ largest when $x=2$

$f'''(x) = -6x^{-4}$

$f''''(x) = 24x^{-5}$

$\frac{24}{x^5}$

largest when $x=2$

$K = \frac{24}{2^5} = \frac{24}{32}$

$K = \frac{3}{4}$

$K = \frac{2}{2^3} = \frac{1}{4}$

$|E_T| \le \frac{1/4(3-2)^3}{12(4)^2} = \boxed{\frac{1}{4(12)(16)}}$

$|E_M| \le \frac{1/4(3-2)^3}{24(4^2)} = \boxed{\frac{1}{4(24)(16)}}$

$|E_S| \le \frac{3/4(3-2)^3}{180(4^4)} = \boxed{\frac{3}{4(180)(4^4)}}$

2. The table shows values of a force function $f(x)$, where x is measured in meters and $f(x)$ in newtons. Use Simpson's Rule and the Trapezoid Rules to estimate the work done by the force in moving an object a distance of 18 m.

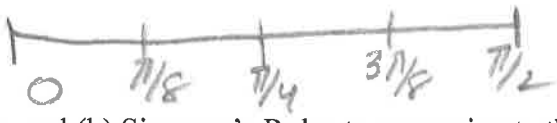
x	0	3	6	9	12	15	18
$f(x)$	9	9	8	8	7	7	7

$n=6$ $\Delta x = 3$

$S_6 = \frac{3}{3}[9+4(9)+2(8)+4(8)+2(7)+4(7)+7]$

$T_6 = \frac{3}{2}[9+2(9)+2(8)+2(8)+2(7)+2(7)+7]$

$\Delta x = \frac{\pi}{8}$



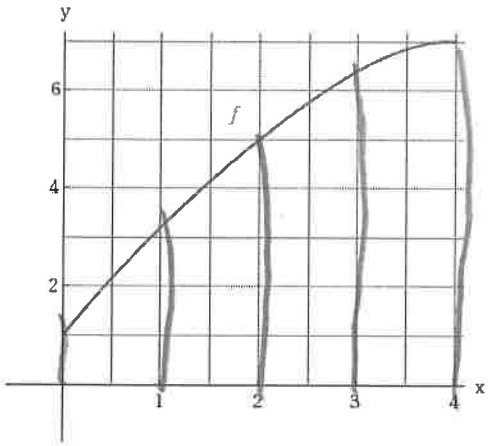
3. Use (a) the Trapezoidal Rule and (b) Simpson's Rules to approximate the given integral with the specified value of n for $\int_0^{\pi/2} \cos 2\theta \, d\theta, \quad n = 4$ Evaluate the trig functions.

$$T_4 = \frac{\pi/8}{2}\left[\cos 0 + 2\cos \pi/4 + 2\cos \pi/2 + 2\cos 3\pi/4 + \cos \pi\right]$$
$$= \frac{\pi}{16}\left[1 + 2(\sqrt{2}/2) + 2(0) + 2(-\sqrt{2}/2) - 1\right]$$
$$= \frac{\pi}{16}\left[1 + \sqrt{2} - \sqrt{2} - 1\right] = \boxed{0}$$

$$S_4 = \frac{\pi/8}{3}\left[\cos 0 + 4\cos \pi/4 + 2\cos \pi/2 + 4\cos 3\pi/4 + \cos \pi\right]$$
$$= \frac{\pi}{24}\left[1 + 4(\sqrt{2}/2) + 2(0) + 4(-\sqrt{2}/2) - 1\right] = \boxed{0}$$

4. $I = \int_0^4 f(x)dx$ where f is the function whose graph is shown.

$\Delta x = 1$



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(a) Use the graph to approximate M_4, L_4, R_4

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$f(x) = x^{1/2}$

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$f'''(x) = \frac{3}{8}x^{-5/2}$

$f''''(x) = \frac{-15}{16}x^{-7/2}$

$\frac{-15}{16x^{7/2}}$ is largest when $x=1$

$K = \frac{15}{16}$

$$\frac{\frac{15}{16}(2-1)^5}{180n^4} \le \frac{1}{10^3}$$

$$\frac{15}{16(180)n^4} \le \frac{1}{10^3}$$

$$\sqrt[4]{\frac{15(10^3)}{16(180)}} \le n$$

the next larger even integer